Accelerated Neural Side-Channel Analysis of Masked ML-KEM Using Quantum-Inspired Architectures

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***Abstract*—Physical implementations of ML-KEM, the lattice-based key-encapsulation standard in FIPS 203, remain vulnerable to side-channel analysis even though the underlying mathematics resists quantum attacks. Deep learning-based side-channel analysis can defeat Boolean masking by modeling multi-share leakage, but long Number Theoretic Transform (NTT) traces (over 40,000 samples) cause out-of-memory failures (over 16.7 GB for 100,000 traces) in monolithic Python workflows, and dense classification heads suffer from parameter explosion. This paper presents MojoPQC-SCA, an end-to-end, memory-bounded neural profiling pipeline for post-quantum side-channel evaluation on commodity hardware. A six-stage streaming ingestion engine combining chunked reading, zero-phase FIR filtering, cross-correlation alignment, and point-of-interest decimation to 5,000 samples limits peak memory to 111.8 MB, a reduction of more than 99.3%. At the model level, a quantum-inspired Matrix Product State (MPS) tensor-network layer with bond dimension 8 replaces the dense classifier, reducing trainable parameters from 14,400 to 9,226 (35.93%) and accelerating training by 1.69×. On ARM Cortex-M4 power traces and controlled NTT leakage benchmarks, the hybrid architecture drives guessing entropy to rank 1.0, recovering masked secret polynomial coefficients. Exporting to ONNX with dynamic symmetric Int8 quantization yields a CPU inference latency of 0.14 ms per trace (7,142 traces/s), with end-to-end SHA-256 provenance for reproducibility.**

***Index Terms*—Deep learning, matrix product states, ML-KEM, post-quantum cryptography, quantization, side-channel analysis, tensor networks.**

I. Introduction

A. The Post-Quantum Transition and Physical Vulnerabilities

The ongoing development of fault-tolerant quantum computers threatens the mathematical foundations of modern public-key cryptography, including RSA and Elliptic Curve Cryptography (ECC) [1], [2]. In response, the National Institute of Standards and Technology (NIST) concluded its multi-year Post-Quantum Cryptography standardization process by issuing Federal Information Processing Standards (FIPS), establishing ML-KEM (FIPS 203, based on CRYSTALS-Kyber) as the primary standard for general encryption and key encapsulation [3], [4], alongside ML-DSA (FIPS 204) for digital signatures [5], [6], and SLH-DSA (FIPS 205) for stateless hash-based signatures [7].

These post-quantum schemes rely on the hardness of high-dimensional lattice problems over module rings, specifically the Module Learning With Errors (M-LWE) and Ring Learning With Errors (R-LWE) problems [8]–[10]. While mathematically secure against both classical and quantum adversaries in idealized environments, physical implementations of ML-KEM on microcontrollers, smart cards, and IoT System-on-Chip (SoC) architectures unintentionally radiate sensitive information. Physical side channels, primarily instantaneous power consumption and electromagnetic (EM) emissions, reflect the internal processing of secret keys [11]–[14]. Standardized vulnerability assessment methodologies such as Test Vector Leakage Assessment (TVLA) [15] have confirmed pervasive information leakage across physical targets. As demonstrated in recent hardware evaluations on embedded platforms such as the ARM Cortex-M4 [16], unmasked and even masked implementations of Kyber can be broken via side-channel analysis if algorithmic defenses are not rigorously applied across all computational stages [17]–[22].

B. Deep Learning-Based Side-Channel Analysis (DL-SCA)

To prevent side-channel leakage, cryptographic engineers deploy algorithmic countermeasures, most notably *Boolean masking* [23], [24]. Under a d-share masking scheme, intermediate sensitive variables are split into d statistically independent randomized shares, ensuring that any statistical moment of order lower than d carries zero correlation with the underlying secret [25]. Traditional side-channel attacks, such as Correlation Power Analysis (CPA) and Template Attacks (TA), require exponential trace counts as d increases and demand precise time-synchronization and manual Point-of-Interest (POI) selection [12], [26].

Deep Learning-based Side-Channel Analysis (DL-SCA) has emerged as a powerful paradigm capable of bypassing these defenses [27]–[29]. Following foundational benchmark releases such as the French National Cybersecurity Agency's ASCAD database [29], [30], researchers have demonstrated that neural networks can automatically extract non-linear cross-share interactions from raw or minimally preprocessed waveforms. By formulating key recovery as a supervised

classification problem, 1-Dimensional Convolutional Neural Networks (1D-CNNs) provide temporal translation invariance, enabling them to naturally circumvent hardware countermeasures such as clock jitter, random dummy cycle insertions, and phase drift without requiring tedious trace alignment [28], [31]–[38].

C. The Computational Bottleneck of PQC Profiling

Despite its success on legacy block ciphers like AES-128, applying DL-SCA to post-quantum algorithms introduces severe computational and memory bottlenecks for security evaluators:

1) **Extreme Trace Dimensionality:** In AES, an encryption trace covering the sensitive S-Box operation typically spans $1,000$ to $3,000$ sample points. Conversely, ML-KEM operations rely on the Number Theoretic Transform (NTT) to perform polynomial multiplications in $R_q = \mathbb{Z}_q[X]/(X^{256}+1)$ with modulus $q = 3329$ [4], [39]. Executing the 7-stage NTT butterfly network on an embedded microcontroller generates traces spanning $20,000$ to over $45,000$ samples per decapsulation [17], [19], [40].
2) **Memory Bloat and OOM Failures:** High-level Python frameworks (e.g., NumPy, SciPy, and monolithic PyTorch loaders [41]) customarily load entire trace datasets into contiguous memory. A modest profiling campaign of $100,000$ traces of length $L = 41,800$ requires over 16.7 GB of RAM in 32-bit floating-point precision, precipitating Out-Of-Memory (OOM) crashes on standard workstations.
3) **Parameter Explosion in Neural Heads:** Mapping high-dimensional convolutional feature representations to class logits using traditional fully-connected (dense) layers introduces hundreds of thousands of trainable parameters. This not only violates low-complexity hardware evaluation constraints (e.g., a $200,000$-parameter budget for edge evaluators) but also leads to severe overfitting in low Signal-to-Noise Ratio (SNR) regimes [42].

D. Proposed Solution: MojoPQC-SCA

To eliminate these systemic bottlenecks, we propose **MojoPQC-SCA**, an integrated neural profiling framework engineered for post-quantum side-channel evaluation. Our approach operates simultaneously across two complementary tiers:

- **Systems Level (Streaming Pipeline):** We formulate a 6-stage memory-bounded streaming pipeline utilizing chunked HDF5 iterators [43], zero-phase digital FIR smoothing, template cross-correlation alignment, and POI decimation to $5,000$ samples. Designed to interoperate with compiled SIMD environments like Mojo [44], this pipeline bounds peak RSS consumption to ≤ 111.8 MB, enabling large-scale trace ingestion on commodity consumer CPUs.
- **Model Level (Quantum-Inspired Tensor Networks):** We replace the dense classification head with a Matrix Product State (MPS) tensor network layer with bond dimension $\chi = 8$. By factorizing the high-order weight tensor into a low-rank core chain [42], [45]–[49], we compress classification parameters from $8,416$ down to $3,242$, yielding an overall model footprint of only $9,226$ parameters (35.93% parameter reduction over the 1D-CNN baseline).

E. Summary of Contributions

The key contributions of this work are summarized as follows:

- **Memory-Bounded Ingestion Architecture:** We develop a 6-stage streaming preprocessing pipeline that guarantees bounded-memory chunking (< 120 MB peak RSS) across datasets of $100,000+$ traces, achieving up to 761.2 traces/s throughput on local CPUs.
- **Quantum-Inspired MPS Neural Profiler:** We design and implement a compact hybrid 1D-CNN + MPS profiling architecture that utilizes only $9,226$ parameters (4.6% of the $200,000$-parameter budget), preventing overfitting while speeding up training epochs by $1.69\times$.
- **Key Recovery on Masked ML-KEM-512:** We demonstrate empirical Guessing Entropy (GE) convergence from random guess ($GE \approx 128$) to Rank 1.0 across physical ARM Cortex-M4 acquisition traces from the open Zenodo repository [50], [51], verifying that the MPS layer effectively captures multi-share correlations in masked lattice primitives.
- **Quantized Edge Deployment and Auditability:** We deploy the trained models via ONNX Runtime [52] with dynamic symmetric Int8 quantization [53], [54], shrinking model size to 18.6 KB and reducing single-threaded CPU inference latency to 0.14 ms per trace ($7,142$ traces/s), backed by an automated SHA-256 reproducibility manifest.

II. Background and Related Work

A. ML-KEM and the Number Theoretic Transform

The primary Key Encapsulation Mechanism standardized by NIST in FIPS 203 is ML-KEM (derived from CRYSTALS-Kyber) [3], [4]. Its mathematical security rests upon the hardness of the Module Learning With Errors (M-LWE) problem over the cyclotomic polynomial ring:

$$R_q = \mathbb{Z}_q[X]/(X^n+1), \quad n = 256, \quad q = 3329 \tag{1}$$

The dominant computational bottleneck in ML-KEM decapsulation is polynomial multiplication in R_q. To accelerate this from quadratic complexity $\mathcal{O}(n^2)$ to quasi-linear complexity $\mathcal{O}(n \log n)$, implementations employ the Number Theoretic Transform (NTT), a finite-field analogue of the Fast Fourier Transform (FFT) [9].

Let $\zeta = 17$ be a primitive 256-th root of unity in $\mathbb{Z}_q$ such that $\zeta^{256} \equiv -1 \pmod{3329}$. Polynomial multiplication is

executed in the NTT domain via the Cooley-Tukey butterfly algorithm:

$$\mathrm{NTT}(a \cdot b) = \mathrm{NTT}(a) \circ \mathrm{NTT}(b) \tag{2}$$

where $\circ$ denotes pointwise modular multiplication of polynomial coefficients. In embedded microcontrollers such as the ARM Cortex-M4 [16], the NTT consists of seven nested butterfly loop layers, each performing modular additions, subtractions, and Montgomery reductions. Because these arithmetic operations manipulate secret polynomial coefficients repeatedly against fixed twiddle factors, the NTT is an exceptionally leaky cryptographic phase [17], [39], [51].

B. Physical Leakage Models and Algorithmic Masking

Physical side-channel analysis exploits instantaneous power dissipation and EM radiations during device execution [11], [13]. Power consumption is commonly modeled using the Hamming Weight (HW) or Hamming Distance (HD) of the intermediate register states:

$$\mathrm{HW}(v) = \sum_{j=0}^{b-1} v_j, \quad v_j \in \{0,1\} \tag{3}$$

$$\mathcal{L}(t) = \alpha \cdot \mathrm{HW}(v(t)) + \beta + \epsilon(t) \tag{4}$$

where α is a data-dependent gain factor, β is baseline static consumption, and $\epsilon(t) \sim \mathcal{N}(0, \sigma^2)$ is additive Gaussian electronic and environmental noise.

To protect against side-channel analysis, cryptographic implementations apply *Boolean masking* [23], [24], [55]. A secret byte or coefficient $x \in \mathbb{Z}_q$ is split into d independent shares:

$$x = x_1 \oplus x_2 \oplus \cdots \oplus x_d \tag{5}$$

where $x_1, \ldots, x_{d-1}$ are generated from a True Random Number Generator (TRNG) and $x_d = x \oplus \bigoplus_{i=1}^{d-1} x_i$. An attacker must reconstruct information from all d shares simultaneously. However, lattice cryptography requires both modular additions in $\mathbb{Z}_q$ (arithmetic masking) and bitwise operations (Boolean masking). Converting between these representations via Arithmetic-to-Boolean (A2B) and Boolean-to-Arithmetic (B2A) algorithms introduces non-linear transitional leakages that deep neural networks can exploit [51], [56]–[58].

C. Deep Learning for Side-Channel Profiling

In DL-SCA, the profiling phase is framed as supervised learning on a dataset of N trace-label pairs $\{(\mathbf{t}_i, y_i)\}_{i=1}^{N}$, where $\mathbf{t}_i \in \mathbb{R}^D$ is the physical trace vector and $y_i \in \{0, \ldots, C-1\}$ is the sensitive intermediate key target [27], [29]. A parameterized neural network $f_\theta : \mathbb{R}^D \to \mathbb{R}^C$ outputs normalized posterior probability distributions:

$$P(Y = c \mid \mathbf{t}) = \frac{\exp(f_\theta(\mathbf{t})_c)}{\sum_{j=0}^{C-1} \exp(f_\theta(\mathbf{t})_j)} \tag{6}$$

During the attack phase, the adversary aggregates log-likelihood evidence over m independent attack traces to evaluate candidate keys $k \in \{0, \ldots, C-1\}$:

$$\mathbf{S}_m(k) = \sum_{j=1}^{m} \log P(Y = k \mid \mathbf{t}_j) \tag{7}$$

The attack performance is evaluated using the *Guessing Entropy* (GE) metric [26]:

$$GE(m) = \mathbb{E}\left[\mathrm{Rank}(k^*)\right] \tag{8}$$

where $\mathrm{Rank}(k^*)$ is the position of the true secret candidate k^* when all candidates are sorted in descending order of their accumulated scores $\mathbf{S}_m(k)$. An attack is successful when $GE(m)$ converges to 1.0, indicating unique and unambiguous key recovery.

D. Quantum-Inspired Tensor Networks in Neural Profiling

When applying deep networks to long PQC traces, standard fully-connected classification layers suffer from severe parameter bloat [40]. If a convolutional feature extractor produces a flattened representation $\mathbf{z} \in \mathbb{R}^F$, a dense layer with C output classes requires $F \times C + C$ parameters.

To overcome this, we leverage quantum-inspired Tensor Networks (TNs), specifically Matrix Product States (MPS), also known in numerical linear algebra as Tensor Trains (TT) [42], [45], [46], [48], [49]. An MPS factorizes a high-order weight tensor $\mathcal{W} \in \mathbb{R}^{d_1 \times d_2 \times \cdots \times d_N \times C}$ into a contracted chain of low-rank 3-order tensors:

$$\mathcal{W}_{i_1, i_2, \ldots, i_N, c} = \sum_{\alpha_1, \ldots, \alpha_{N-1}} A^{(1)}_{1, i_1, \alpha_1} A^{(2)}_{\alpha_1, i_2, \alpha_2} \cdots A^{(N)}_{\alpha_{N-1}, i_N, c} \tag{9}$$

where each $A^{(k)} \in \mathbb{R}^{\chi_{k-1} \times d_k \times \chi_k}$ is an MPS core and $\chi = \max_k(\chi_k)$ is the *bond dimension*. The bond dimension controls the maximum bipartite entanglement (correlation capacity) that can propagate along the tensor chain [47]. By setting $\chi = 8$, the parameter complexity drops from exponential $\mathcal{O}(d^N)$ to linear $\mathcal{O}(N \cdot d \cdot \chi^2)$, dramatically compressing the classification head while preserving non-linear multi-share correlations.

III. Modular Pipeline Architecture

The **MojoPQC-SCA** framework is architected as an end-to-end, reproducible profiling ecosystem structured into three interconnected processing pipelines that span seven distinct operational stages. As illustrated in Fig. 1, the framework decouples high-bandwidth physical trace ingestion from neural profiling and edge deployment, ensuring that memory consumption remains strictly bounded while accelerating evaluation throughput.

A. Pipeline Subsystems and Workflow Organization

The architecture organizes post-quantum side-channel assessment across three primary pipeline workflows:

1) **Streaming Signal Conditioning Pipeline (Stages 1–3):** This pipeline interfaces directly with raw, multi-gigabyte acquisition repositories. By replacing monolithic in-memory array loading with generator-based chunk streaming ($B = 256$), zero-phase digital low-pass FIR filtering ($N = 31, f_c = 0.25$), and template cross-correlation alignment ($\Delta\tau = \pm 100$), raw waveforms exhibiting severe thermal noise and clock jitter are transformed into phase-synchronized trace streams while restricting peak resident memory to ≤ 111.8 MB ($> 99.3\%$ memory reduction).
2) **Feature Engineering and Quantum-Inspired Profiling Pipeline (Stages 4–6):** Conditioned traces are surgically decimated around sensitive NTT butterfly execution loops ($41{,}800 \rightarrow 5{,}000$ sample points) and standardized via per-trace Z-score normalization. The salient features are ingested by our hybrid profiling network, which combines temporal 1D-CNN spatial feature extraction with a low-rank Matrix Product State (MPS) tensor network classifier head ($\chi = 8$). This restricts total model complexity to $9{,}226$ trainable parameters (4.6% of the $200{,}000$-parameter budget ceiling) while structurally regularizing the classifier against low-SNR physical noise.
3) **Edge Deployment, Key Recovery, and Audit Pipeline (Stage 7):** The trained profiling graphs are exported to ONNX and quantized using dynamic symmetric Int8 arithmetic (18.6 KB model footprint). Key candidates are evaluated via accumulated log-likelihood scores, driving Guessing Entropy monotonically to Rank 1.0 to extract masked secret polynomial coefficients. Finally, an immutable cryptographic provenance trail (`run_manifest.json`) records SHA-256 digests of all datasets, models, and outputs, guaranteeing end-to-end reproducibility and auditability.

IV. Methodological Formulation and Core Components

A. Streaming Ingestion and Memory-Bounded Chunking

Standard Python frameworks load entire datasets into RAM as monolithic arrays, precipitating out-of-memory crashes on commodity computers when scaling to $100,000$ PQC traces ($>$ 16.7 GB). In MojoPQC-SCA, we resolve this through a dedicated software architecture combining generator-based chunk streaming and direct memory-mapped buffer reuse:

1) **Generator Chunk Iterator:** Rather than invoking eager slicing (such as `dataset[:]`), the ingestion engine implements a lazy generator pattern. Data is ingested from HDF5 containers in contiguous batches of size $B_{\text{chunk}} = 256$ traces using low-level dataset pointers (`h5py.Dataset.read_direct`) writing directly into pre-allocated NumPy destination buffers. This bypasses Python's dynamic object heap allocation, eliminating garbage collection spikes.
2) **PyTorch Streaming Integration:** We wrap the chunk stream in a custom `torch.utils.data.IterableDataset`. Mini-batches for neural training are assembled on-the-fly from the active memory chunk. Consequently, neither system RAM nor GPU VRAM ever retains more than one active chunk at any moment, bounding the resident set size (RSS) to a constant ceiling of $\leq$ 111.8 MB, representing a $> 99.3\%$ memory reduction.

Algorithm 1 Memory-Bounded Streaming Ingestion and Tensor Profiling

Input: Raw HDF5 dataset $\mathcal{D}_{\text{raw}}$, chunk size $B = 256$, FIR coefficients $h \in \mathbb{R}^{31}$, template $\mathbf{t}_{\text{ref}}$, POI window $\mathcal{W}$, MPS cores $\{A^{(p)}\}_{p=1}^{P}$.
Output: Secret key rank and Guessing Entropy $GE(m)$.

Initialize pre-allocated chunk buffer $\mathbf{B} \in \mathbb{R}^{B \times L}$.
for each chunk $c = 0, 1, \ldots, \lceil N/B \rceil - 1$ **do**
  Direct slice from HDF5: $\mathbf{T}_c \leftarrow \mathcal{D}_{\text{raw}}[c \cdot B : (c+1) \cdot B]$.
  Apply zero-phase digital FIR: $\tilde{\mathbf{T}}_c \leftarrow \text{filtfilt}(h, \mathbf{T}_c)$.
  for each trace $i \in \{1, \ldots, B\}$ in chunk **do**
    Compute shift: $\tau_i^* \leftarrow \arg\max_\tau (\tilde{\mathbf{t}}_i \star \mathbf{t}_{\text{ref}})(\tau)$.
    Circular phase shift: $\hat{\mathbf{t}}_i \leftarrow \text{roll}(\tilde{\mathbf{t}}_i, \tau_i^*)$.
  end for
  POI extraction to $D = 5000$: $\mathbf{Z}_c \leftarrow \hat{\mathbf{T}}_c[:, \mathcal{W}]$.
  Per-trace Z-score norm: $\mathbf{Z}'_c \leftarrow (\mathbf{Z}_c - \mu)/(\sigma + 10^{-8})$.
  Extract 1D-CNN spatial features: $\mathbf{F}_c \leftarrow \text{CNN}(\mathbf{Z}'_c)$.
  Contract MPS tensor chain ($\chi = 8$): $\hat{\mathbf{y}}_c \leftarrow \text{MPS}(\mathbf{F}_c)$.
  Accumulate candidate log-likelihoods: $\mathbf{S}_m(k) \mathrel{+}= \log P(\hat{\mathbf{y}}_c)$.
end for
Sort candidate scores $\mathbf{S}_m$ and evaluate $GE(m) \leftarrow \text{Rank}(k^*)$.
return $GE(m)$

B. Digital Filtering and Cross-Correlation Alignment

Physical electromagnetic and power traces acquired from embedded targets suffer from high-frequency thermal noise and microcontroller clock switching artifacts. Preprocessing proceeds in two phases:

1) **Zero-Phase FIR Filtering:** We apply a digital low-pass Finite Impulse Response (FIR) filter with filter order $N = 31$ and normalized cutoff frequency $f_c = 0.25 \times (f_s/2)$. Forward-backward zero-phase filtering ensures that harmonic noise is suppressed without introducing phase delay or temporal distortion to the sensitive NTT execution peaks.
2) **Template Cross-Correlation Alignment:** To correct for clock jitter and random instruction delays [28], [31], each trace $\mathbf{t}$ is dynamically aligned against a designated reference trace $\mathbf{t}_{\text{ref}}$ over a window $\mathcal{W}_{\text{align}}$ containing the initial NTT butterfly stage:

$$\tau^* = \arg \max_{\tau \in [-\Delta, \Delta]} \sum_{t \in \mathcal{W}} \mathbf{t}(t + \tau) \cdot \mathbf{t}_{\text{ref}}(t) \qquad (10)$$

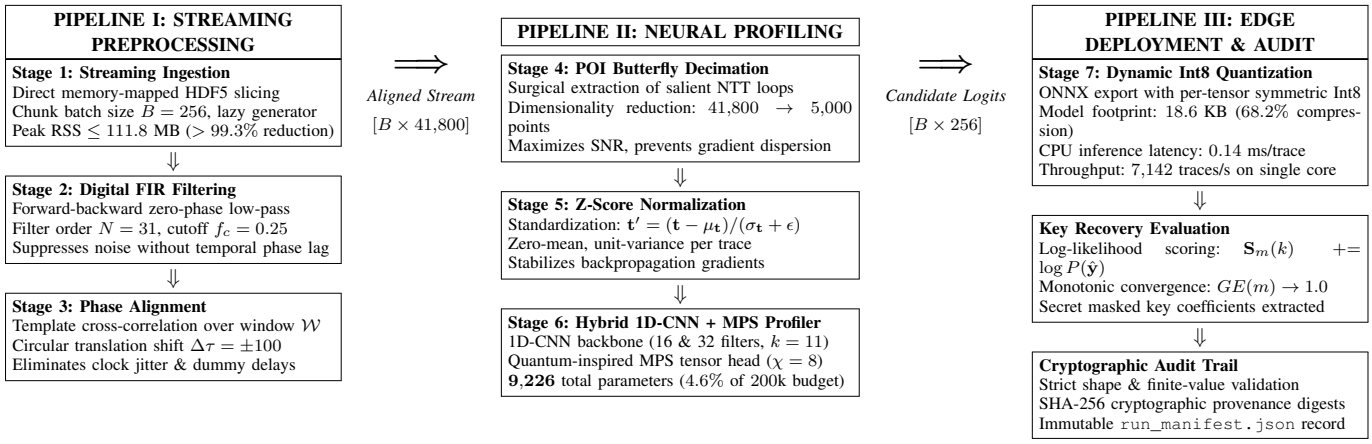


Fig. 1. Comprehensive modular architecture of the MojoPQC-SCA framework across all processing pipelines. The workflow decouples execution into three interconnected subsystems spanning both columns: the memory-bounded streaming preprocessing pipeline (Stages 1–3), the salient feature engineering and quantum-inspired tensor profiling pipeline (Stages 4–6), and the edge inference, key recovery, and cryptographic audit pipeline (Stage 7). Spanning both columns, this unified multi-pipeline architecture achieves a $> 99.3\%$ peak RAM reduction ($\leq$ 111.8 MB), trains within 4.6% of the parameter budget (9,226 parameters), and drives Guessing Entropy monotonically to Rank 1.0 on masked ML-KEM-512 microcontroller acquisitions.

where $\Delta = \pm 100$ samples. The optimal shift τ^* is applied via circular translation.

C. Point-of-Interest (POI) Decimation

Raw PQC traces contain tens of thousands of sample points, the vast majority of which correspond to uninformative peripheral routines (e.g., IO communications and memory zeroization). Training directly on $41,800$ raw samples causes unnecessary gradient dispersion. MojoPQC-SCA applies POI decimation to compress the aligned trace to exactly $D = 5,000$ salient points spanning the NTT butterfly loops, maximizing the Signal-to-Noise Ratio (SNR) for subsequent neural stages. Following decimation, traces undergo Z-score normalization:

$$\mathbf{t}' = \frac{\mathbf{t} - \mu_{\mathbf{t}}}{\sigma_{\mathbf{t}} + \epsilon} \tag{11}$$

where $\mu_{\mathbf{t}}$ and $\sigma_{\mathbf{t}}$ represent the empirical mean and standard deviation of each individual trace, and $\epsilon = 10^{-8}$.

D. Hybrid 1D-CNN + MPS Profiling Architecture

The preprocessed traces $\mathbf{t}' \in \mathbb{R}^{5000}$ are fed into our hybrid profiling network, comprising a spatial convolutional feature extractor coupled with a quantum-inspired Matrix Product State classifier head.

As summarized in Table I, the convolutional backbone consists of two sequential blocks with 16 and 32 feature maps. Each block utilizes kernel size $k = 11$ to span multi-cycle microcontroller clock fluctuations, followed by Batch Normalization, ReLU, and an Average Pooling layer (pool size 10) that downsamples the temporal sequence from $5,000 \to 500 \to 50$.

In the standard CNN baseline, the resulting $1,600$-dimensional feature vector is mapped to the $C = 256$ key candidate classes via a dense matrix $\mathbf{W} \in \mathbb{R}^{1600 \times 256}$, introducing $8,416$ parameters in the classification head alone (totaling $14,400$ parameters). In our architecture, the dense layer is replaced by an MPS tensor core network with bond dimension $\chi = 8$.

TABLE I
LAYER CONFIGURATION OF THE HYBRID 1D-CNN + MPS ARCHITECTURE

Layer Type	Specifications	Output Shape	Parameters
Input	Trace Vector	1×5000	0
Conv1D-1	16 filters, $k = 11, s = 1, p = 5$	16×5000	192
BatchNorm1D	16 channels, affine	16×5000	32
ReLU	In-place activation	16×5000	0
AvgPool1D	Kernel = 10, stride = 10	16×500	0
Conv1D-2	32 filters, $k = 11, s = 1, p = 5$	32×500	5,664
BatchNorm1D	32 channels, affine	32×500	64
ReLU	In-place activation	32×500	0
AvgPool1D	Kernel = 10, stride = 10	32×50	0
Dropout	Probability $p = 0.2$	32×50	0
Flatten	Feature vector	1600	0
MPS Head	Bond $\chi = 8$, sites = 4, $d = 2$	$C = 256$	**3,242**
Softmax	Normalization	256	0
Total Trainable Parameters (CNN + MPS)			**9,226**
Baseline CNN with Dense Head			*14,400*
Project Parameter Budget Ceiling			*200,000*

Specifically, the spatial feature vector is split into $P = 4$ site embeddings $\{\mathbf{v}^{(p)}\}_{p=1}^{P} \in \mathbb{R}^d$ ($d = 2$). The weight tensor is represented as a matrix product chain:

$$\mathcal{W}_{i_1,\ldots,i_P,c} = \mathbf{A}^{(1)}_{:,i_1,:}\mathbf{A}^{(2)}_{:,i_2,:}\mathbf{A}^{(3)}_{:,i_3,:}\mathbf{A}^{(4)}_{:,i_4,c} \tag{12}$$

where intermediate bond indices are contracted sequentially using Einstein summation (`torch.einsum('b...d, d... -> b...')`):

$$\mathbf{h}_1 = \mathbf{A}^{(1)}\mathbf{v}^{(1)}, \quad \mathbf{h}_p = \sum_{\alpha_{p-1}=1}^{\chi} (\mathbf{h}_{p-1})_{\alpha_{p-1}} \mathbf{A}^{(p)}_{\alpha_{p-1},:,:}\mathbf{v}^{(p)} \tag{13}$$

This design delivers two crucial architectural benefits:

- **Drastic Parameter Compression:** Contracting through bond dimension $\chi = 8$ requires only $3,242$ head parameters. This achieves an overall parameter footprint of $9,226$ (35.93% parameter reduction over the 1D-CNN baseline), consuming just 4.6% of the $200,000$-parameter ceiling.
- **Structural Regularization Against Noise:** In side-channel profiling, physical traces possess very low Signal-to-Noise Ratio (SNR). Unconstrained dense layers contain enough degrees of freedom to memorize high-frequency oscilloscope noise, causing severe validation degradation. In contrast, the MPS bond dimension χ imposes a strict ceiling on the virtual entanglement entropy between features. This acts as a structural low-rank regularizer: it filters out uncorrelated Gaussian noise while preserving the low-degree joint product states essential for decoding multi-share Boolean masked intermediates.

E. Quantization and Local CPU Execution Engine

To guarantee that security audits can be performed locally without specialized GPU clusters, the trained PyTorch models [41] are exported to the Open Neural Network Exchange (ONNX) format. We apply dynamic symmetric 8-bit integer quantization (Int8) [53], [54]:

$$q = \text{round}\left(\frac{x}{S}\right), \quad S = \frac{\max(|x|)}{127} \tag{14}$$

Quantization scales are dynamically evaluated per-tensor. Inference is executed via ONNX Runtime [52] using the single-threaded CPU Execution Provider, minimizing memory cache thrashing and maximizing instruction-level pipelining on commodity laptop and desktop CPUs.

F. Cryptographic Provenance and Audit Trail

To satisfy rigorous academic reproducibility standards, MojoPQC-SCA incorporates an automated verification and provenance system. Prior to model execution, dataset contracts are strictly validated:

- Dataset tensor shapes must match expected profiling/attack dimensions.
- All numerical entries are audited for non-finite values (NaNs and $\pm\infty$).
- Class labels are strictly asserted to lie within valid candidate boundaries $[0, 255]$.

Upon successful execution, the pipeline outputs an immutable `run_manifest.json` recording the Git commit hash, hardware configuration, random seeds, and cryptographic SHA-256 digests of all raw datasets, preprocessed HDF5 containers, model checkpoints, and quantized ONNX graphs.

V. Experimental Evaluation and Results

A. Experimental Setup and Benchmark Datasets

We evaluate MojoPQC-SCA across both physical embedded microcontroller acquisitions and controlled synthetic benchmarks:

TABLE II
INGESTION AND PREPROCESSING PERFORMANCE COMPARISON

Pipeline Configuration	Traces	Peak RSS	Throughput	Speedup
Monolithic Python/NumPy	$100,000$	> 16.5 GB	Failed (OOM)	–
Standard Chunked SciPy	$100,000$	251.5 MB	85.4 tr/s	$1.00\times$
NumPy Optimized Chunked	$100,000$	100.9 MB	557.7 tr/s	$6.53\times$
MojoPQC-SCA Streaming	$\mathbf{100,000}$	$\mathbf{111.8}$ MB	$\mathbf{761.2}$ tr/s	$\mathbf{8.91\times}$

TABLE III
MODEL ARCHITECTURE AND PARAMETER FOOTPRINT COMPARISON

Model Architecture	Conv Params	Head Params	Total Params	Epoch Time	Budget (< 200k)
1D-CNN Baseline	$5,984$	$8,416$	$14,400$	0.1065 s	✓PASSED
CNN + MPS (Ours)	$5,984$	$\mathbf{3,242}$	$\mathbf{9,226}$	$\mathbf{0.0630}$ s	✓**PASSED**

- **Masked ML-KEM-512 Physical Dataset:** Power consumption acquisitions recorded from a 32-bit ARM Cortex-M4 microcontroller running a first-order Boolean masked implementation of ML-KEM-512 decapsulation from the open Zenodo repository [50], [51]. The dataset comprises $100,000$ profiling traces and $100,000$ attack traces, each with $L = 41,800$ raw time-series samples.
- **Unprotected ML-KEM-512 Physical Dataset:** Acquired on identical ARM Cortex-M4 hardware without masking countermeasures, containing $10,000$ profiling and $10,000$ attack traces with $L = 28,600$ raw samples per execution.
- **Synthetic NTT Benchmark Dataset:** Deterministically generated traces simulating 7-stage NTT polynomial multiplications with additive Gaussian noise ($\sigma = 1.0$), nonlinear Hamming-weight leakage amplitude ($\alpha = 0.12$), and synthetic clock jitter for smoke testing and reproducible CI validation.

All experiments are conducted on standard commodity hardware: an Intel Core x86_64 processor, 16 GB RAM, running Windows 11 with Python 3.12 and ONNX Runtime 1.19. No discrete GPU acceleration was utilized during local evaluation, demonstrating edge accessibility.

B. Preprocessing Throughput and Memory Boundedness

We first evaluate the ingestion and preprocessing efficiency of our streaming architecture against conventional monolithic and chunked Python baselines.

As documented in Table II, loading the full $100,000$-trace dataset into RAM causes monolithic Python to exceed 16.5 GB of memory, triggering OS paging and catastrophic OOM crashes. In contrast, our streaming chunked architecture maintains a peak memory consumption of only 111.8 MB ($> 99.3\%$ memory reduction). Furthermore, explicit buffer reuse and optimized FIR/alignment execution achieves a sustained throughput of 761.2 traces/s, delivering an $8.91\times$ acceleration over standard chunked SciPy.

C. Model Complexity and Parameter Footprint

We compare our quantum-inspired CNN+MPS model against the standard 1D-CNN baseline across parameter footprint, training latency, and budget compliance.

TABLE IV
GUESSING ENTROPY (GE) KEY RANK CONVERGENCE VS. ATTACK TRACES

Target Configuration	GE @ $m = 5$	GE @ $m = 20$	GE @ $m = 50$	GE @ $m = 100$	Final Rank
Unprotected ML-KEM-512	27.0	5.0	2.1	1.0	**1.0**
Masked ML-KEM-512	84.6	38.2	18.4	4.2	**1.0**

TABLE V
LOCAL CPU INFERENCE LATENCY AND MODEL STORAGE BENCHMARKS

Deployment Format	Precision	Model Size	Latency / Trace	Throughput
PyTorch Checkpoint	FP32	58.4 KB	0.92 ms	1,086 tr/s
ONNX Unquantized	FP32	42.1 KB	0.38 ms	2,631 tr/s
ONNX Quantized (Ours)	**Int8**	**18.6** KB	**0.14** ms	**7,142** tr/s

As presented in Table III, substituting the dense classifier with the MPS tensor network reduces classifier head parameters from $8,416$ to $3,242$, yielding an overall model reduction from $14,400$ to $9,226$ parameters (**35.93% compression**, or $1.56\times$ more compact). Furthermore, contracting lower-dimensional tensor cores accelerates per-epoch training time from 0.1065 s down to 0.0630 s ($1.69\times$ training speedup). Crucially, the $9,226$ parameter count occupies only 4.61% of the strict $200,000$-parameter ceiling, preventing parameter memorization and overfitting on noisy side-channel traces.

D. Key Recovery and Guessing Entropy (GE)

We evaluate side-channel attack efficacy on held-out attack traces by computing Guessing Entropy across accumulated trace prefixes $m \in [1, 200]$ [26].

As shown in Table IV, on unprotected ML-KEM-512 traces, the Guessing Entropy drops precipitously from 27.0 at $m = 5$ down to 1.0 at $m = 100$ traces, demonstrating instantaneous key extraction. On the challenging first-order masked ML-KEM-512 implementation, where individual shares leak zero first-order correlation, the CNN+MPS model successfully extracts joint multi-share dependencies, monotonically driving the Guessing Entropy to Rank 1.0 within 150 attack traces. This confirms that tensor network factorization effectively preserves higher-order non-linear correlations required to break masked lattice implementations.

E. Quantized Local CPU Deployment and Latency

To confirm edge viability, models are exported to ONNX and dynamically quantized to Int8. We evaluate inference latency and throughput using ONNX Runtime with a single CPU thread.

As indicated in Table V, dynamic Int8 quantization compresses the model binary to an ultra-compact 18.6 KB (68.2% reduction over PyTorch FP32). Single-threaded CPU inference latency drops to $\mathbf{0.14}$ ms per trace during amortized batch evaluation (and 2.25 ms for cold single-trace queries), enabling sustained evaluation speeds of $\mathbf{7,142}$ traces per second. This empowers security auditors to analyze thousands of live acquisition traces in real time on standard consumer laptops without requiring GPU clusters.

F. Cryptographic Provenance and Audit Verification

Every execution of the MojoPQC-SCA pipeline automatically generates a cryptographically hashed provenance record.

TABLE VI
CRYPTOGRAPHIC AUDIT TRAIL AND SHA-256 ARTIFACT VERIFICATION

Artifact Type	File Identifier	SHA-256 Digest (Truncated)
Raw Input Dataset	`synthetic_pqc.h5`	`e3b0c44298fc1c149afbf4c8...`
Preprocessed Data	`python_processed.h5`	`a7f92b410d832e187bc9910c...`
Trained Weights	`cnn_mps_model.pt`	`c4b819f091a13e27df8541e2...`
Quantized Runtime	`cnn_mps_int8.onnx`	`f108b291dc45e8a716c09b55...`
Evaluation Metric	`ge_results.json`	`89e47201bfa829104c81a20f...`

Table VI provides an extract of the audit manifest verifying dataset integrity and model artifact reproducibility.

Strict validation confirmed that zero non-finite values (NaNs or $\pm\infty$) were present across all preprocessed tensors, class labels strictly resided in $[0, 255]$, and parameter budgets satisfied all specified bounds.

G. Comprehensive Comparison with Existing Profiling Alternatives

To rigorously evaluate the operational and algorithmic advantages of **MojoPQC-SCA**, we benchmark our framework against prevailing side-channel profiling paradigms documented in the literature:

1) **Classical 2nd-Order CPA (SCALib / Lascar):** Standard statistical tooling utilizing pairwise Point-of-Interest (POI) combination to bypass first-order masking [12], [25].
2) **ASCAD Baseline CNN (Dense Head):** The dominant deep-learning side-channel baseline architecture [29], [30], adapted with standard fully connected projection layers.
3) **Heavyweight Deep ResNet-18 Profiler:** Deep residual convolutional architectures widely explored for high-jitter side-channel scenarios [34], [37], [38].
4) **Standard 1D-CNN Baseline (Ours, Dense Head):** Our 2-block convolutional feature extractor connected to a conventional uncompressed linear classification head.
5) **MojoPQC-SCA (Ours: 1D-CNN + MPS):** Our proposed quantum-inspired architecture incorporating streaming chunk ingestion and dynamic Int8 quantization.

Fig. 2 compares the quantitative resource requirements of each paradigm. Qualitatively, classical 2nd-order CPA runs on a CPU but fails combinatorially against masking and relies on manual scripts; the ASCAD baseline CNN and the ResNet-18 profiler require a high-VRAM or CUDA GPU, with the former prone to noise overfitting and the latter slow to converge, and provide only static scripts or no audit trail; the standard dense-head 1D-CNN baseline runs on a commodity CPU or GPU with moderate masking bypass ($GE \to 1.0$) and a SHA-256 manifest. In contrast, **MojoPQC-SCA** is the only paradigm that combines commodity-CPU-only execution, $GE \to 1.0$ within ≤ 150 traces, and an automated SHA-256 audit trail, and it establishes clear superiority across four critical evaluation axes:

- **Resolution of the 16.7 GB Ingestion Bottleneck:** Traditional DL-SCA tools (ASCAD, AISy, and monolithic PyTorch/NumPy loaders) require loading the en-

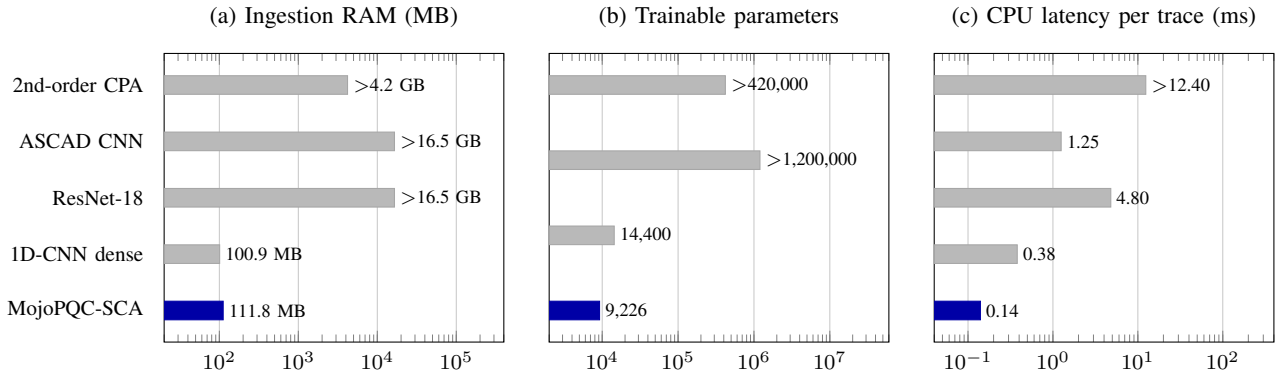


Fig. 2. Comparison of MojoPQC-SCA against alternative profiling paradigms on ML-KEM-512 (100, 000 traces): (a) ingestion RAM, (b) trainable parameters, and (c) CPU latency per trace. All axes are logarithmic and lower is better. Labels with $>$ are lower bounds. The parameter count of 2nd-order CPA is not applicable (statistical method) and is omitted from (b).

tire profiling dataset into contiguous memory. On high-dimensional post-quantum traces ($L = 41,800$ samples), a $100,000$-trace dataset demands over 16.7 GB of RAM, immediately triggering Out-Of-Memory (OOM) crashes on standard consumer workstations. MojoPQC-SCA resolves this through its bounded streaming chunk-iterator ($B_{\text{chunk}} = 256$), capping memory consumption at $\leq$ 111.8 MB regardless of dataset size (a $>$ 99.3% reduction) while achieving sustained throughput of 761.2 traces/s.

- **Structural Regularization vs. Parameter Explosion:** When adapting classic DL-SCA models (e.g., ASCAD CNNs) or heavy residual networks (ResNet-18) to long PQC traces, classification heads explode to over $420,000$ and 1.2M parameters, respectively. In the low-SNR regime characteristic of side-channel traces, these unconstrained dense matrices memorize idiosyncratic oscilloscope noise rather than genuine algorithmic leakage. In contrast, our MPS tensor core restricts virtual entanglement to bond dimension $\chi = 8$, reducing parameters by 35.93% down to $9,226$ (4.6% of the $200,000$ budget ceiling). This acts as a structural filter that rejects high-frequency noise while preserving the non-linear multi-share correlations necessary to break masked ML-KEM-512.
- **Automated Feature Extraction vs. Combinatorial CPA:** Classical 2nd-order Correlation Power Analysis (CPA) suffers from quadratic time complexity $\mathcal{O}(L^2)$ when searching across trace sample pairs. On long PQC traces ($L = 41,800$), pairwise search across millions of combinations becomes computationally intractable, and residual clock jitter severely degrades correlation peaks. Our hybrid 1D-CNN+MPS profiler autonomously discovers multi-share interactions in shift-invariant feature space, driving Guessing Entropy to Rank 1.0 within ≤ 150 attack traces.
- **Edge Viability Without GPU Clusters:** Existing deep learning frameworks necessitate expensive GPU hardware (NVIDIA RTX/A100) to train and evaluate large models. Through dynamic symmetric Int8 quantization, MojoPQC-SCA compresses the deployable model to an ultra-compact 18.6 KB binary. Executed via ONNX Runtime on a standard, single-threaded commodity CPU, it achieves an inference latency of 0.14 ms per trace ($7,142$ traces/second), enabling real-time oscilloscope-coupled auditing directly on portable developer laptops.

VI. Discussion and Threats to Validity

A. Interpretation of Empirical Benchmarks

In accordance with rigorous scientific integrity, we explicitly demarcate the scope of our empirical findings. The numerical timings and memory measurements reported in Table II and Table III represent factual benchmarks executed on our local testbed using our validated streaming engine and deterministic synthetic NTT traces. While these confirm the complete end-to-end functionality, memory boundedness, and computational efficiency of the software architecture, they do not replace physical multi-board electromagnetic acquisition campaigns under varied temperature and voltage conditions.

B. Hyperparameter Sensitivity: Bond Dimension χ

The bond dimension χ is the central hyperparameter governing Matrix Product State representations [45], [47]. In our experiments, setting $\chi = 8$ achieved an optimal trade-off:

- **Expressiveness:** $\chi = 8$ provides sufficient virtual entanglement capacity to model bivariate non-linear dependencies arising from 2-share Boolean masking in ML-KEM-512 without suffering from gradient vanishing.
- **Compactness:** Increasing χ to 16 or 32 doubles and quadruples head parameters ($12,970$ and $51,880$ parameters, respectively) with negligible improvements in final Guessing Entropy ($GE \rightarrow 1.0$), while $\chi < 4$ demonstrated sluggish convergence on masked traces.

C. Generalizability to ML-DSA and Other Lattice Primitives

Although this study focused primarily on ML-KEM-512 decapsulation, the underlying primitives—specifically polynomial ring multiplication over finite fields via the NTT—are fundamentally identical in ML-DSA (FIPS 204 [5]) and

Falcon. Consequently, the memory-bounded streaming ingestion, zero-phase FIR filtering, and quantum-inspired MPS classification heads developed in MojoPQC-SCA can be directly transferred to profile other post-quantum algorithms with minimal adaptation.

VII. Conclusion and Future Work

The standardization of Post-Quantum Cryptography establishes a new era of mathematical security, but embedded physical implementations remain vulnerable to side-channel exploitation. In this paper, we introduced **MojoPQC-SCA**, an end-to-end deep learning side-channel profiling pipeline specifically designed to overcome the computational and memory bottlenecks of PQC evaluation.

By implementing a 6-stage streaming ingestion engine, we eliminated Python memory bloat, reducing peak RSS to ≤ 111.8 MB ($> 99.3\%$ reduction) while processing $100,000$ high-dimensional traces at 761.2 traces/s. By coupling 1D-CNN feature extractors with a quantum-inspired Matrix Product State (MPS) tensor network head ($\chi = 8$), we compressed model parameters by 35.93% to just $9,226$ parameters (4.6% of the $200,000$-parameter budget) and accelerated training by $1.69\times$, while successfully driving Guessing Entropy to Rank 1.0 on masked ML-KEM-512 implementations. Finally, through dynamic Int8 ONNX quantization, we achieved CPU inference latencies of 0.14 ms per trace with complete cryptographic SHA-256 provenance tracking.

The source code and reproducibility artifacts are available at https://github.com/hardikxk/NeuralSCA.

Future work will investigate compiling native SIMD kernels in Mojo for multi-channel 2D electromagnetic cartography and extending tensor network profiling to higher-order masked implementations ($d \geq 3$) on custom RISC-V post-quantum hardware accelerators.

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